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A UNIFORM STRONG-COUPLING SPECTRAL GAP AND A CENTER SELECTION RULE FOR THE SU(N) KOGUT–SUSSKIND LATTICE HAMILTONIAN

Main result: for $g \geq g_0(N)$, the SU(N) Kogut–Susskind Hamiltonian on every open cubic box has a unique, gauge-invariant ground state and spectral gap at least $g^2 c_N/4$, uniformly in the volume. Any expansion term that reaches a fundamental Wilson loop on an $r \times s$ rectangle carries at least rs distinct plaquettes; given a convergent center-covariant expansion, this yields an area law with explicit string tension. No continuum statement is made.

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Abstract

We prove a volume-uniform spectral gap for the open-boundary SU(N) Kogut–Susskind lattice Hamiltonian at strong coupling: $H_L - E_{0,L} \geq (g^2 c_N/4)(\mathbb{1} - P_{0,L})$ for $g \geq g_0(N)$, with $c_N = (N^2 - 1)/N$. The proof embeds link variables as sites of an enlarged lattice and applies Yarotsky's gap theorem for weak perturbations of non-interacting quantum lattice systems, which allows infinite-dimensional on-site spaces, unbounded on-site operators and non-translation-invariant interactions. We prove an exact center selection rule: in any expansion around the electric vacuum, a term contributes to the expectation of a fundamental Wilson loop on a planar $r \times s$ rectangle only if it contains at least rs distinct plaquettes. Combined with a lattice-animal count, this gives a Wilson-loop area law with explicit constants whenever a uniformly convergent center-covariant expansion exists. We also record a ground-state transform, a curvature criterion needed only in horizontal directions, and a Schur-test reduction of that criterion to a row-sum estimate. We do not prove a continuum mass gap, nor that a static area bound implies a gap at arbitrary coupling. Definitions, statements and proofs are written in the mathematical English customary in Physical Review.

Provenance

This white paper is the formal edition of the strong-coupling paper of 6 November 2026, arranged in the conventions of the companion paper B8 [10]. Its numbering maps to the earlier draft as follows.

Here	Earlier draft	Change
Definitions 1–11	§2, §4.1, §5.1	Collected and numbered
Lemma 1: normalization and constants	§2, “Constants”	Stated and proved as a lemma
Lemma 2: ground-state transform	Lemma 3.1	—
Proposition 3: horizontal curvature criterion	Proposition 3.2	—
Theorem 4: uniform strong-coupling gap	Theorem 4.1	—
Lemma 5: center selection rule	Lemma 5.1	—
Proposition 6: area count	Proposition 5.2	—
Corollary 7: expansion terms	Corollary 5.3	—
Proposition 8: conditional area law	§6, Steps one–three	New here; from [8, Lemma 1(ii)]
Proposition 9: Schur-test criterion	§6, curvature route	New here; formalizes the row-sum remark
Remarks 10–12	§7 remarks	Entanglement remark updated for [10]

0 Definitions

Definition 1 (gauge group). Fix $N \geq 2$ and let $G = SU(N)$ with fundamental representation ρ and the bi-invariant metric $\langle X, Y\rangle = -\mathrm{Tr}(XY)$ on $\mathfrak{su}(N)$. We write $c_N = (N^2 - 1)/N$ and $B(X, Y) = 2N\,\mathrm{Tr}(XY)$ for the Killing form.

Definition 2 (lattice and configuration space). Λ_L is the open cubic complex with vertices $\{0, \dots, L\}^3$ and unit spacing, with positively oriented links E_L and one orientation per elementary plaquette P_L. The configuration manifold is $\mathscr{M}_L = G^{E_L}$ with the product bi-invariant metric and normalized product Haar measure μ_L. Link reversal is $U_{\bar{e}} = U_e^{-1}$, U_p is the ordered plaquette holonomy, and for a closed path C, $W(C) = N^{-1}\,\mathrm{Tr}\,\rho(\Pi_{e\in C} U_e)$.

Definition 3 (gauge transformations). For $h \in G^{\Lambda_L}$, $(\mathscr{U}(h)F)(U) = F((h_{s(e)} U_e h_{t(e)}^{-1})_e)$. The physical space $\mathscr{H}_L^{\mathrm{phys}}$ is the subspace of $L^2(\mathscr{M}_L)$ fixed by all $\mathscr{U}(h)$.

Definition 4 (Hamiltonian and spectral data). The Kogut–Susskind Hamiltonian is

$$H_L = (g^2/2)\,\Sigma_{e\in E_L}(-\Delta_e) + g^{-2}\,\Sigma_{p\in P_L}[N - \mathrm{ReTr}\,\rho(U_p)],$$

with ground-state energy $E_{0,L}$, normalized ground state Ω_L, $A_L = H_L - E_{0,L}\mathbb{1}$, $P_{0,L} = |\Omega_L\rangle\langle\Omega_L|$ and $Q_L = \mathbb{1} - P_{0,L}$.

Definition 5 (normalized Hamiltonian). $h_e = -\Delta_e/c_N$, $\Psi_p = -(2/(g^4 c_N))\,\mathrm{ReTr}\,\rho(U_p)$, and $\bar{H}_L = \Sigma_e h_e + \Sigma_p \Psi_p$. $\Omega_0 = \mathbb{1}$ denotes the electric vacuum.

Definition 6 (vacuum measure). $\pi_L = \Omega_L^2\,\mu_L$, $h_L = \log \Omega_L$ and $D_L f = \Delta f + 2\langle \nabla h_L, \nabla f\rangle$. A tangent vector of $\mathscr{M}_L$ is horizontal if it is orthogonal to the gauge orbit through its base point.

Definition 7 (link-site embedding). $\iota(x, x + e_i) = 2x + e_i \in \mathbb{Z}^3$, $\Lambda_0 = \{0, 1, 2\}^3$, $B_L = \{0, \dots, 2L + 2\}^3$, $\varphi_{2x} = \Sigma_p \Psi_p$ with lower corner x, and $\varphi_z = 0$ for $z \notin 2\mathbb{Z}^3$.

Definition 8 (Yarotsky constants) [2, Thm 1]. $c_1, c_2 > 0$ are constants for range Λ_0 such that every Hamiltonian $\Sigma_{z\in\Lambda} h_z + \Sigma_{z+\Lambda_0\subset\Lambda} \varphi_z$, with each $h_z \geq 0$ self-adjoint with nondegenerate kernel and gap 1 and each φ_z bounded, self-adjoint, supported in $z + \Lambda_0$ with $\sup_z \|\varphi_z\| < c_1$, has a nondegenerate ground state and gap at least $1 - c_2 \sup_z \|\varphi_z\|$, on every finite Λ. We set $\beta^* = \min\{c_1/2, (2c_2)^{-1}\}$ and $g_0(N) = (6N/(c_N\beta^*))^{1/4}$.

Definition 9 (center transformations). $\zeta = e^{2\pi i/N}$. For $\eta \in \mathbb{Z}_N^{E_L}$, $(T_\eta F)(U) = F((\zeta^{\eta_e} U_e)_e)$, $(\delta\eta)(p) = \Sigma_{e\in\partial p} \pm\eta_e$ and $\eta(C) = \Sigma_{e\in C} \pm\eta_e$, with signs given by orientation.

Definition 10 (expansion hypothesis $\mathfrak{E}(g, K)$). $\langle\Omega_L, W(C)\Omega_L\rangle = \Sigma_{X\in\mathscr{P}(C)} w_X(C)$, where $\mathscr{P}(C)$ is the set of connected plaquette sets containing a plaquette that contains a link of C; each w_X is a finite sum of matrix elements $\langle F\Omega_0, W(C)F'\Omega_0\rangle$ with F, F' built from $\{\Psi_p\}_{p\in X}$ and functions of $\Sigma_e(-\Delta_e)$; and $|w_X| \leq (Kg^{-4})^{|X|}$, uniformly in L.

Definition 11 (polymer Hessian row sum). For a representation $\log \Omega_L = \Sigma_X \varphi_X$ with C^2 terms, $r_L = \sup_e \Sigma_X \Sigma_{f\in E_L} \|\mathrm{Hess}_{ef}\,\varphi_X\|_\infty$, where Hess_{ef} is the (e, f) block of the Hessian in the product metric.

1 Normalization and the ground-state transform

Lemma 1 (normalization and constants). (i) $\mathrm{Ric}_G = (N/2)\langle\cdot,\cdot\rangle$, hence $\mathrm{Ric}_{\mathscr{M}_L} \geq N/2$ on every tangent vector.

(ii) $c_N = (N^2-1)/N$ is the smallest positive eigenvalue of $-\Delta_G$; for SU(2), $c_2 = 3/2$.

(iii) $\bar{H}_L = 2H_L/(g^2 c_N) - 2N|P_L|/(g^4 c_N)\cdot\mathbb{1}$; each h_e has nondegenerate kernel and gap 1; $\|\Psi_p\| \leq 2N/(g^4 c_N)$ and $\sup_e \Sigma_{p\ni e}\|\Psi_p\| \leq 8N/(g^4 c_N)$.

Proof. (i) For a bi-invariant metric, $\mathrm{Ric}(X,X) = -\tfrac{1}{4}B(X,X) = -(N/2)\mathrm{Tr}\,X^2 = (N/2)|X|^2$. On the product manifold the Ricci tensor is the direct sum of the factors' Ricci tensors.

(ii) $-\Delta_G$ acts on the matrix coefficients of the irreducible representation of highest weight λ by the Casimir $C(\lambda) = (\lambda, \lambda + 2\varrho)$, in the normalization where long roots have squared length 2. In fundamental-weight coordinates $C(\lambda) = \Sigma_{ij}\,\lambda_i (A^{-1})_{ij}(\lambda_j + 2)$, with $(A^{-1})_{ij} = \min(i,j) - ij/N > 0$. Both terms are increasing in every λ_i, so the minimum over $\lambda \neq 0$ is attained at a fundamental weight ω_k, where $\Sigma_j (A^{-1})_{kj} = k(N-k)/2$ gives $C(\omega_k) = k(N-k)/N + k(N-k) = k(N-k)(N+1)/N$. This is minimal at $k \in \{1, N-1\}$, with value $(N^2-1)/N$. For $N = 2$ it equals $3/2$, the first eigenvalue on S^3 of radius $\sqrt{2}$.

(iii) The identity follows by expanding Definition 5. By (ii), h_e has the constant function as its unique ground state and gap exactly 1. Since $|\mathrm{ReTr}\,\rho(U)| \leq N$, $\|\Psi_p\| \leq 2N/(g^4 c_N)$. A link lies in the two planes that contain it and in at most two plaquettes in each, hence in at most four plaquettes, including at the open boundary. ∎

Lemma 2 (ground-state transform). For smooth f on $\mathscr{M}_L$,

$$\Omega_L^{-1} A_L(\Omega_L f) = -(g^2/2)\, D_L f, \quad \langle \Omega_L f, A_L(\Omega_L f)\rangle = (g^2/2)\int |\nabla f|^2 \, d\pi_L .$$

Consequently, a uniform Poincaré inequality $\mathrm{Var}_{\pi_L}(f) \leq C_P \int |\nabla f|^2 d\pi_L$ for gauge-invariant f is equivalent to a physical gap of at least $g^2/(2C_P)$.

Proof. Expanding $\Delta(\Omega_L f) = f\Delta\Omega_L + 2\langle\nabla\Omega_L, \nabla f\rangle + \Omega_L \Delta f$, the terms proportional to f cancel by $H_L\Omega_L = E_{0,L}\Omega_L$, which gives the first identity. Since $D_L = \Omega_L^{-2}\,\mathrm{div}(\Omega_L^{\,2}\nabla)$, integration by parts on the compact manifold gives the second. Multiplication by Ω_L is unitary from $L^2(\pi_L)$ onto $L^2(\mu_L)$ and maps constants to Ω_L, so the spectral gap of A_L on physical states is $(g^2/2)$ times the best Poincaré constant's reciprocal on gauge-invariant functions. ∎

The imaginary-time correlation of the earlier area-to-gap analysis is the stationary autocovariance of the reversible diffusion generated by $(g^2/2)D_L$.

2 Horizontal curvature criterion

Proposition 3 (horizontal curvature criterion). Suppose there is $\kappa > 0$, independent of L, such that for every L and every horizontal vector v,

$$(\mathrm{Ric}_{\mathscr{M}_L} - 2\,\mathrm{Hess}\, h_L)(v, v) \geq \kappa |v|^2.$$

Then $A_L \geq (g^2\kappa/2)\, Q_L$ on physical states.

Proof. The integrated weighted Bochner identity reads

$$\int (D_L f)^2 \, d\pi_L = \int [\, |\mathrm{Hess}\, f|^2 + (\mathrm{Ric} - 2\,\mathrm{Hess}\, h_L)(\nabla f, \nabla f)\,]\, d\pi_L .$$

D_L commutes with the gauge action, so its spectrum on physical states is spanned by gauge-invariant eigenfunctions. For such f, df vanishes on vectors tangent to gauge orbits, so ∇f is horizontal everywhere and the hypothesis applies. For an eigenfunction with $-D_L f = \lambda f$ and $\lambda > 0$ this gives $\lambda^2 \|f\|^2 \geq \kappa\lambda \|f\|^2$, so $\lambda \geq \kappa$, and Lemma 2 restores the factor $g^2/2$. The argument stays on the smooth product manifold, so singular gauge orbits cause no difficulty. ∎

By Lemma 1(i), a sufficient condition is $\mathrm{Hess}\, \log \Omega_L \leq b$ on horizontal vectors with $2b < N/2$. This is the Bakry–Émery method [3, 4]. Shen, Zhu and Zhu [7] apply the same method to the Euclidean lattice Yang–Mills measure at strong coupling: for $SU(N)$ with $\beta < 1/(16(d-1))$ in 't Hooft scaling, in any dimension $d > 1$, they prove a unique infinite-volume measure, log-Sobolev and Poincaré inequalities and exponential decay of correlations. Proposition 3 is the Hamiltonian counterpart, applied to the ground-state measure π_L rather than the Euclidean Gibbs measure.

3 Uniform strong-coupling gap

We use Theorem 1 of Yarotsky [2] and its gap estimate (5). The hypotheses as stated in the source, and how the construction meets them:

Hypothesis of [2, Thm 1]	Construction here	Met
On-site spaces, possibly infinite-dimensional	$L^2(G)$ at every site	Yes
$h_x \geq 0$ self-adjoint, possibly unbounded, unique ground state	$h_z = -\Delta_G/c_N$	Yes
Normalized on-site gap 1	Lemma 1(ii)	Yes
Interactions bounded, self-adjoint, supported on $\Lambda_0 + x$	φ_{2x}, $\Lambda_0 = \{0,1,2\}^3$	Yes
Smallness $\sup_x \|\varphi_x\| < c_1$	$6N/(g^4 c_N) \leq \beta^* < c_1$	For $g \geq g_0(N)$
Finite volume, terms with $\Lambda_0 + x \subset \Lambda$, empty boundary condition	The padded box B_L contains every support	Yes
Translation invariance	Not assumed in Theorem 1	Not needed

Theorem 4 (uniform strong-coupling gap). For every $g \geq g_0(N)$ and every $L \geq 1$, H_L has a unique ground state, which is physical, and

$$H_L - E_{0,L}\,\mathbb{1} \geq (g^2 c_N/4)(\mathbb{1} - P_{0,L}),$$

uniformly in L, on the full tensor product and hence on physical states.

Proof. *Links as sites.* The k-th coordinate of $2x + e_i$ is odd exactly when $k = i$, so ι is injective. Every site of B_L carries $L^2(G)$ and $h_z = -\Delta_G/c_N$.

Plaquette supports and interactions. A plaquette in directions i, j with lower corner x has link sites $2x + e_i$, $2x + e_j$, $2x + 2e_i + e_j$ and $2x + e_i + 2e_j$, all in $2x + \Lambda_0$. At most three plaquettes share a lower corner, so by Lemma 1(iii) $\sup_z \|\varphi_z\| \leq \beta(g) = 6N/(g^4 c_N)$, and each φ_{2x} is self-adjoint.

Padding. Plaquettes of Λ_L have lower-corner coordinates at most $L - 1$ in their two directions, so the four link sites $\iota(\partial p)$ lie in $\{0, \ldots, 2L\}^3$. The support box $2x + \Lambda_0$ extends two units beyond $2x$ in every direction, including the one normal to the plaquette, where $2x$ can reach $2L$; hence $2x + \Lambda_0 \subset \{0, \ldots, 2L + 2\}^3 = B_L$, boundary-face plaquettes included, and the padding is needed.

Applying [2]. For $g \geq g_0(N)$, $\beta(g) \leq \beta^* < c_1$, so the enlarged Hamiltonian $H^{\mathrm{enl}} = \Sigma_{z \in B_L} h_z + \Sigma_z \varphi_z$ has a nondegenerate ground state and normalized gap at least $1 - c_2\beta^* \geq \tfrac{1}{2}$, with constants independent of L.

Decoupling. With $J_L = \iota(E_L)$ and $R_L = B_L \setminus J_L$, every interaction acts on J_L only, so $H^{\mathrm{enl}} = \bar{H}_L \otimes \mathbb{1} + \mathbb{1} \otimes \Sigma_{z \in R_L} h_z$. The second summand has a unique product ground state and gap 1. For decoupled systems with unique ground states the gap of the sum is the smaller gap, so $\tfrac{1}{2} \le \min\{\mathrm{gap}(\bar{H}_L), 1\}$ and $\mathrm{gap}(\bar{H}_L) \ge \tfrac{1}{2}$.

Undoing the normalization. By Lemma 1(iii), $H_L = (g^2 c_N/2)\bar{H}_L + \mathrm{const}\cdot\mathbb{1}$, so $\mathrm{gap}(H_L) \ge g^2 c_N/4$ on the full tensor product.

Gauge descent. The ground state is unique and, since e^{-tH_L} is positivity improving, strictly positive. Gauge transformations commute with H_L and map positive normalized states to positive normalized states, so they fix Ω_L, which is therefore physical. Restricting the form inequality to $\mathscr{H}_L^{\mathrm{phys}}$ preserves the bound. ∎

The threshold $g_0(N)$ is expressed through the constants c_1, c_2 of [2]; its reduction to a single certified number and its N-uniformity are proved in [10, Theorem 1]. The gap is in lattice units and grows like g^2, so it carries no information about the continuum limit $g \to 0$. The proof uses neither Proposition 3 nor any Wilson-loop bound.

4 Center selection rule and area count

Lemma 5 (center selection rule). Let X be a set of plaquettes, and let F, F' be built from $\{\Psi_p : p \in X\}$ and functions of $\Sigma_e(-\Delta_e)$. If C is not the boundary of a $\mathbb{Z}_N$-valued 2-chain supported in X, then $\langle F\Omega_0,\, W(C)F'\Omega_0\rangle = 0$.

Proof. We first record four properties of T_η. It is unitary on $L^2(\mathscr{M}_L)$ and fixes Ω_0. It commutes with every $-\Delta_e$, because multiplication by a central element is an isometry of the bi-invariant metric, and therefore with every function of $\Sigma_e(-\Delta_e)$, including resolvents. It satisfies $T_\eta \Psi_p T_\eta^{-1} = \Psi_p$ whenever $(\delta\eta)(p) \equiv 0 \pmod N$. Finally, $T_\eta W(C) = \zeta^{\eta(C)} W(C)$.

Let K be the span of $\{\partial p : p \in X\}$ in $M = \mathbb{Z}_N^{E_L}$. By hypothesis $C \notin K$, so $[C] \neq 0$ in the finite abelian group M/K, and some character χ of M/K has $\chi([C]) \neq 1$. Since N annihilates M/K, χ takes values in the N-th roots of unity, and every such character of $\mathbb{Z}_N^{E_L}$ has the form $\chi(m) = \zeta^{\langle \eta,\, m\rangle}$ for some η. Triviality on K gives $(\delta\eta)(p) \equiv 0$ for all $p \in X$, while $\eta(C) \not\equiv 0 \pmod N$. Every ingredient of F and F' then commutes with T_η, so $T_\eta F\Omega_0 = F\Omega_0$ and $T_\eta F'\Omega_0 = F'\Omega_0$. By unitarity,

$$\langle F\Omega_0,\, W\, F'\Omega_0\rangle = \langle T_\eta F\Omega_0,\, T_\eta (W\, F'\Omega_0)\rangle = \zeta^{\eta(C)} \langle F\Omega_0,\, W\, F'\Omega_0\rangle,$$

and since $\zeta^{\eta(C)} \neq 1$ the inner product vanishes. ∎

Proposition 6 (area count). Every $\mathbb{Z}_N$-valued 2-chain Σ on the open cubic lattice with $\partial\Sigma = C$, for a planar $r \times s$ rectangle C, has at least rs plaquettes in its support.

Proof. Orthogonal projection π onto the plane of C is a cellular map of the cubic complex, so π_* commutes with ∂; plaquettes perpendicular to the plane go to zero and perpendicular links go to points. Hence $\pi_*\Sigma$ is a 2-chain in a planar box with $\partial(\pi_*\Sigma) = C$. A planar box has no 3-cells and trivial second homology, so a 2-chain there is determined by its boundary: $\pi_*\Sigma$ is the filled rectangle, with coefficient 1 on each of the rs unit squares. The coefficient of each square is the sum of Σ's coefficients on the plaquettes stacked above it. Each such sum is nonzero, so Σ has at least one plaquette above every square. ∎

Corollary 7 (expansion terms). In any Duhamel or Rayleigh–Schrödinger expansion of $\langle \Omega_L,\, W(C)\Omega_L\rangle$ around Ω_0 whose terms are built from the Ψ_p and functions of the electric Hamiltonian, every nonzero term uses a plaquette set X with $|X| \geq rs$. The same holds after regrouping into a linked-cluster or connected expansion, including after cancellation of vacuum contributions.

Proof. Lemma 5 kills every term whose plaquette set carries no $\mathbb{Z}_N$ 2-chain with boundary C, and Proposition 6 bounds the support of any such chain below by rs. A plaquette inserted several times counts once in X, so multiplicities and orientations are covered. A term of a regrouped expansion is a finite sum of original terms and inherits the property. Convergence of the regrouped expansion is a separate matter, addressed by hypothesis $\mathfrak{C}$ in Proposition 8. ∎

5 Conditional area law and the curvature route

Proposition 8 (conditional area law). Assume $\mathfrak{E}(g, K)$ and $q = 12eKg^{-4} \le \tfrac{1}{2}$. Then for every L and every planar $r \times s$ rectangle $C \subset \Lambda_L$,

$$|\langle \Omega_L, W(C)\Omega_L \rangle| \le M\, e^{-\sigma rs}, \quad \sigma = \tfrac{1}{2} \ln q^{-1}, \quad M = 32/(e\sigma).$$

Proof. By Corollary 7, $w_X = 0$ unless X carries a $\mathbb{Z}_N$ 2-chain Σ with $\partial\Sigma = C$, and then $|X| \ge rs$. Every link of C has nonzero coefficient in $\partial\Sigma$, so X contains a plaquette containing a link of C; there are at most $4 \cdot 2(r + s)$ such plaquettes. In the plaquette adjacency graph, where two plaquettes are adjacent when they share a link, each plaquette has at most $4 \cdot 3 = 12$ neighbours, so the number of connected sets of size n containing a given plaquette is at most $(12e)^n$. Hence

$$|\langle W(C) \rangle| \le \Sigma_{n \ge rs}\, 8(r + s)(12e)^n (Kg^{-4})^n = 8(r + s)\Sigma_{n \ge rs} q^n \le 16(r + s)q^{rs}.$$

Since $r + s \le 2rs$ and $xq^{x/2} \le 2/(e \ln q^{-1})$ for $x \ge 0$, $16(r + s)q^{rs} \le 32rs\, q^{rs/2} \cdot q^{rs/2} \le (64/(e \ln q^{-1}))q^{rs/2} = Me^{-\sigma rs}$. All bounds are uniform in L. ∎

Hypothesis $\mathfrak{E}$ combines Step one (a uniformly convergent expansion with vacuum normalization handled) and the summability needed in Step three of the earlier draft. One route: Yarotsky's proof of [2, Thm 1] constructs the ground state; if it writes Ω_L, or the ground-state projector, as a convergent sum of local pieces built from the φ's and on-site resolvents, every piece satisfies the hypotheses of Lemma 5 and his bounds would supply $\mathfrak{E}$.

Proposition 9 (Schur-test criterion). Suppose $\log \Omega_L = \Sigma_X \varphi_X$ with $\sup_L r_L \le b$ and $2b < N/2$. Then $A_L \ge (g^2/2)(N/2 - 2b)Q_L$ on physical states, uniformly in L.

Proof. $\mathrm{Hess}\, h_L = \Sigma_X \mathrm{Hess}\, \varphi_X$ is a symmetric operator on the tangent space, with (e, f) blocks $H_{ef} = \Sigma_X \mathrm{Hess}_{ef} \varphi_X$. By the Schur test for symmetric block matrices, its operator norm is at most $\sup_e \Sigma_f \|H_{ef}\| \le r_L \le b$. With Lemma 1(i), $(\mathrm{Ric} - 2\,\mathrm{Hess}\, h_L)(v, v) \ge (N/2 - 2b)|v|^2$ for every v, in particular for horizontal v, and Proposition 3 applies with $\kappa = N/2 - 2b > 0$. ∎

A C^2 bound on each polymer separately does not suffice: the row sum must be bounded after summing over all polymers touching a link, with polymer-counting and support-size factors included. A first-order computation suggests $g^4 \gtrsim 32/(N^2 - 1)$, but without a remainder bound this is not a rigorous threshold. Theorem 4 is an independent benchmark for any such expansion.

Remarks

Remark 10 (static hypotheses). The full ground state determines H_L, because the kinetic term is fixed and $V_L = E_{0,L} + (g^2/2)\Delta\Omega_L/\Omega_L$, and so it determines the gap. Partial static data such as Wilson-loop averages do not obviously do so. A spectator construction, with a double-well weight on one corner plaquette and Haar measure elsewhere, satisfies a uniform area bound while its gap vanishes as L grows; but it is the ground state of a modified potential. It shows only that any proof that an area bound implies a gap must use the specific plaquette potential.

Remark 11 (first-order transitions). If a bulk first-order transition occurs at some coupling, coexisting confining phases could satisfy an area bound while tunnelling closes finite-volume gaps. Whether the Hamiltonian formulation has such a transition, and how open boundaries affect the near-degeneracy, is not known to us.

Remark 12 (entanglement). At strong coupling the vacuum is a small perturbation of the unentangled product state Ω_0, which is the structure Yarotsky's theorem exploits. The companion paper [10] turns this into theorems: a Lieb–Robinson bound with unbounded electric terms, exponential clustering of vacuum correlations by adaptation of the Nachtergaele–Sims argument, an $O(g^{-4})$ bound on the electric energy of every link, and an entanglement area law up to a polylogarithmic factor under an explicit energy-controlled locality hypothesis. At weak coupling the vacuum is expected to be entangled across all scales, one way to state why the continuum limit lies outside the methods used here.

6 Status

Result	Status	Remaining input
Lemma 1: normalization and constants	Proved	—
Lemma 2: ground-state transform	Proved	—
Proposition 3: horizontal curvature criterion	Proved	Its hypothesis is not verified for Ω_L
Theorem 4: uniform strong-coupling gap	Proved, via [2]	Numerical value of the Yarotsky constants; reduced to one constant κ_Y in [10]
Lemma 5, Proposition 6, Corollary 7: selection rule and area count	Proved	—
Proposition 8: Wilson-loop area law	Conditional	Hypothesis $\mathfrak{C}(g, K)$
Proposition 9: Schur-test criterion	Conditional	A polymer representation with $\sup_L r_L \le b < N/4$
Area bound implies gap at arbitrary fixed g	Open	Remark 10
Continuum mass gap ($g \to 0$)	Not addressed	See [8]

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GitHub repository: https://github.com/teknoquants/STRONG-COUPLING-SPECTRAL-GAP-FOR-THE-SU-N-KOGUT-SUSSKIND-LATTICE-HAMILTONIAN
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